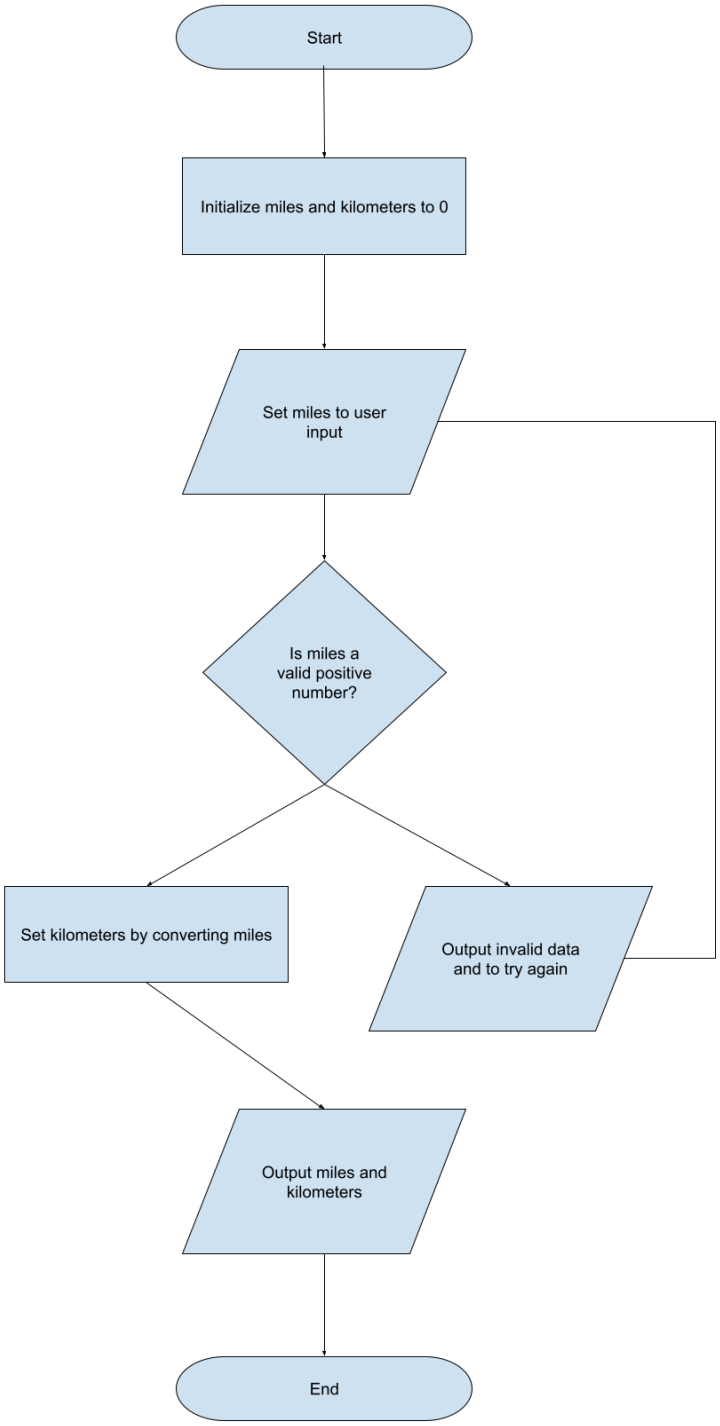
Jeff Victorino
04/05/2026
Module 2.2 Assignment

Flowchart:

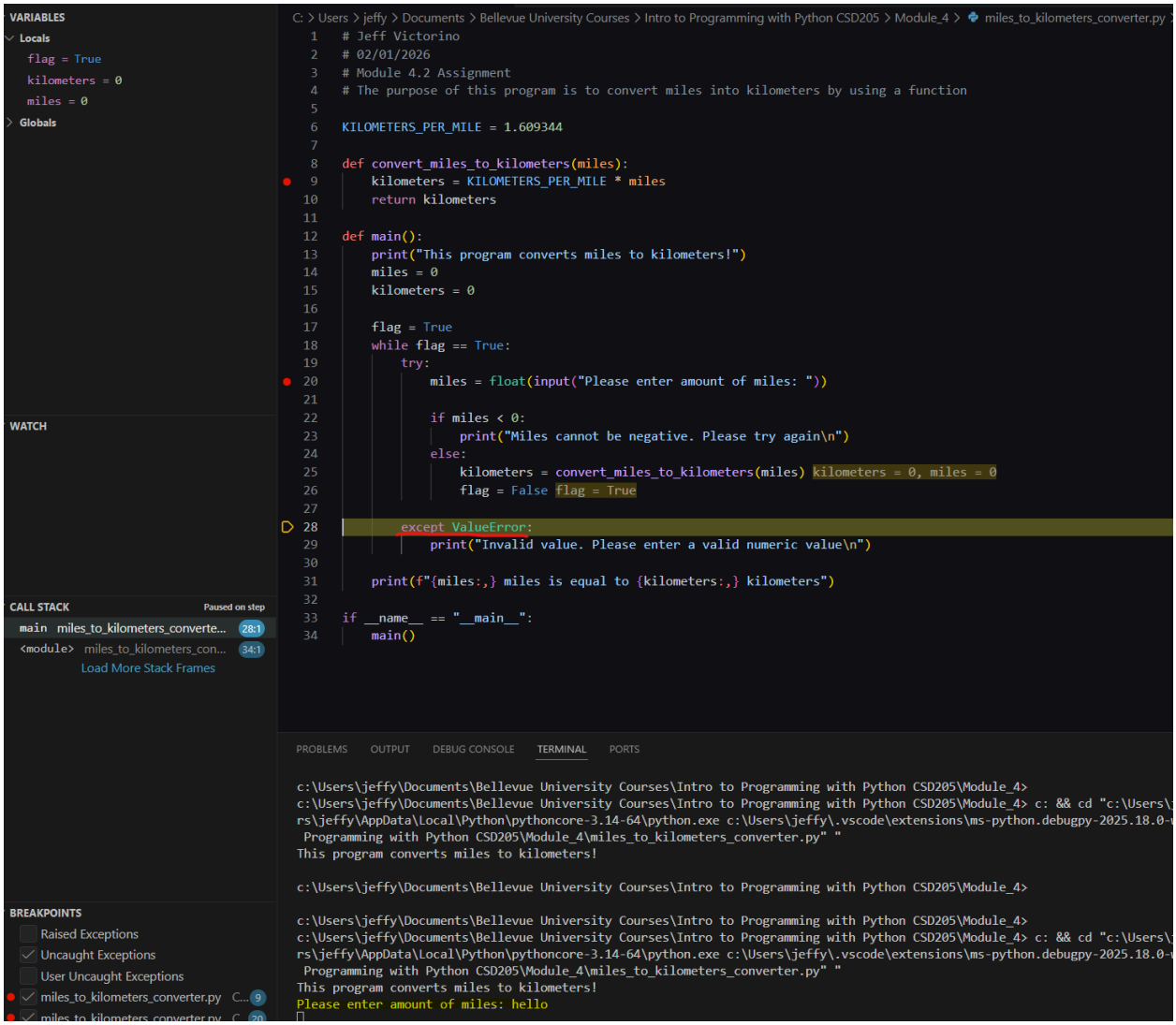


Debugging:

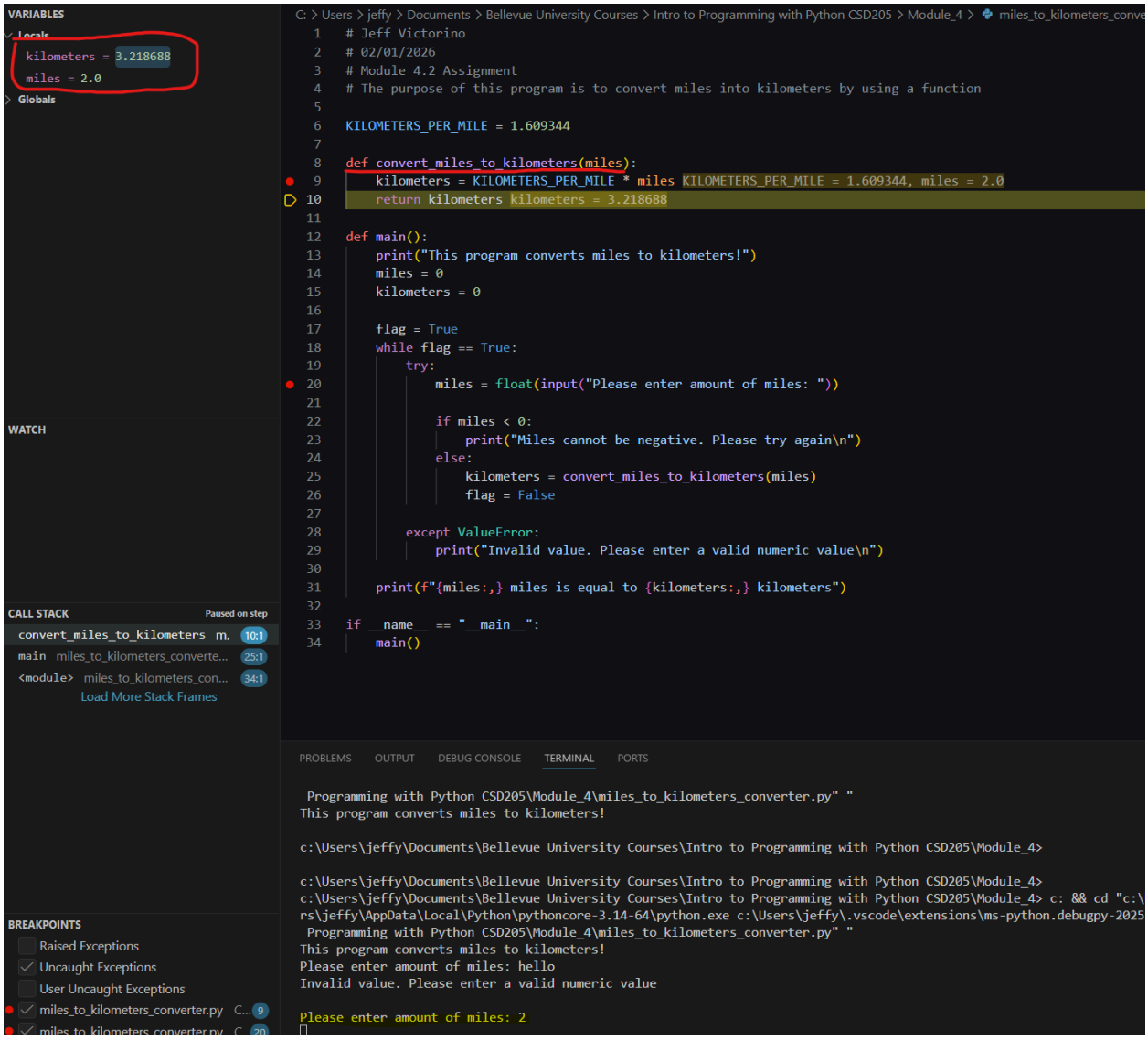
Two breakpoints, one in the main function and one in the conversion function

```python
KILOMETERS_PER_MILE = 1.609344

def convert_miles_to_kilometers(miles):
    kilometers = KILOMETERS_PER_MILE * miles
    return kilometers

def main():
    print("This program converts miles to kilometers!")
    miles = 0
    kilometers = 0

    flag = True
    while flag == True:
        try:
            miles = float(input("Please enter amount of miles: "))

            if miles < 0:
                print("Miles cannot be negative. Please try again\n")
            else:
                kilometers = convert_miles_to_kilometers(miles)
                flag = False

        except ValueError:
            print("Invalid value. Please enter a valid numeric value\n")

    print(f"{miles:,} miles is equal to {kilometers:,} kilometers")

if __name__ == "__main__":
    main()
```

The breakpoint in the main function will allow us to see if miles is valid data (a positive number). If it isn't a positive number, the program asks for input again.



If the input is correct then we reach the second breakpoint in the conversion function. At this point we are certain that miles is valid input.



The program ends by printing the miles and kilometers correctly.

VARIABLES

Locals

special variables

function variables

KILOMETERS_PER_MILE = 1.609344

(return) main = None

Globals

WATCH

CALL STACK Paused on step

<module> miles_to_kilometers_con... 34:1

BREAKPOINTS

Raised Exceptions

Uncaught Exceptions

User Uncaught Exceptions

miles_to_kilometers_converter.py C... 9

miles_to_kilometers_converter.py C... 20

C: > Users > jeffy > Documents > Bellevue University Courses > Intro to Programming with Python CSD205 > Module_4 > miles_to

```python
# Jeff Victorino
# 02/01/2026
# Module 4.2 Assignment
# The purpose of this program is to convert miles into kilometers by using a function

KILOMETERS_PER_MILE = 1.609344 KILOMETERS_PER_MILE = 1.609344

def convert_miles_to_kilometers(miles):
    kilometers = KILOMETERS_PER_MILE * miles
    return kilometers

def main():
    print("This program converts miles to kilometers!")
    miles = 0
    kilometers = 0

    flag = True
    while flag == True:
        try:
            miles = float(input("Please enter amount of miles: "))

            if miles < 0:
                print("Miles cannot be negative. Please try again\n")
            else:
                kilometers = convert_miles_to_kilometers(miles)
                flag = False

        except ValueError:
            print("Invalid value. Please enter a valid numeric value\n")

    print(f"{miles:,} miles is equal to {kilometers:,} kilometers")

if __name__ == "__main__":
    main()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
This program converts miles to kilometers!

c:\Users\jeffy\Documents\Bellevue University Courses\Intro to Programming with Python CSD205\Module_4>

c:\Users\jeffy\Documents\Bellevue University Courses\Intro to Programming with Python CSD205\Module_4>
c:\Users\jeffy\Documents\Bellevue University Courses\Intro to Programming with Python CSD205\Module_4>
rs\jeffy\AppData\Local\Python\pythoncore-3.14-64\python.exe c:\Users\jeffy\.vscode\extensions\ms-pytho
 Programming with Python CSD205\Module_4\miles_to_kilometers_converter.py" "
This program converts miles to kilometers!
Please enter amount of miles: hello
Invalid value. Please enter a valid numeric value

Please enter amount of miles: 2
2.0 miles is equal to 3.218688 kilometers
```